

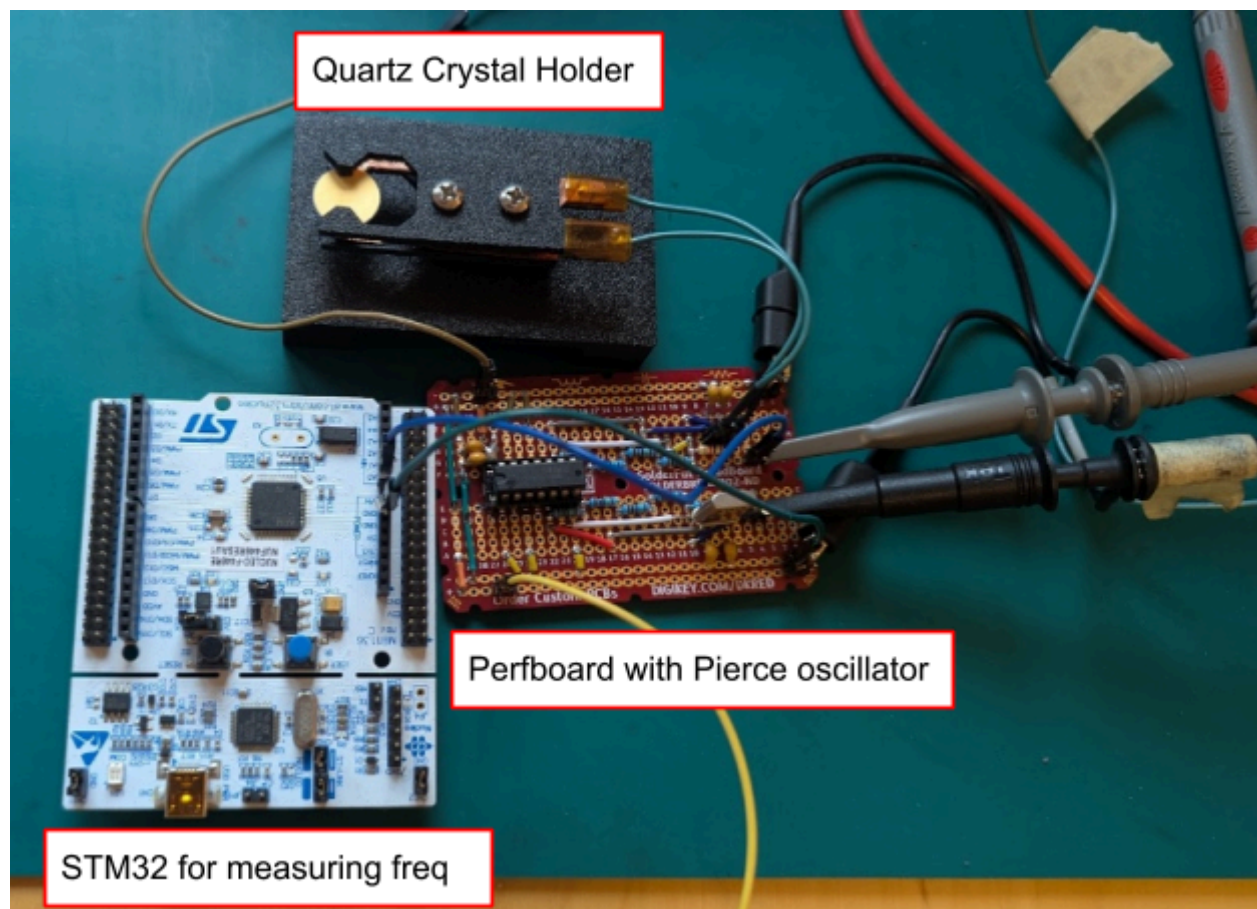
Radu Dehel - Portfolio

Quartz Crystal Microbalance (Hacker Fab team) In Progress

Project Description

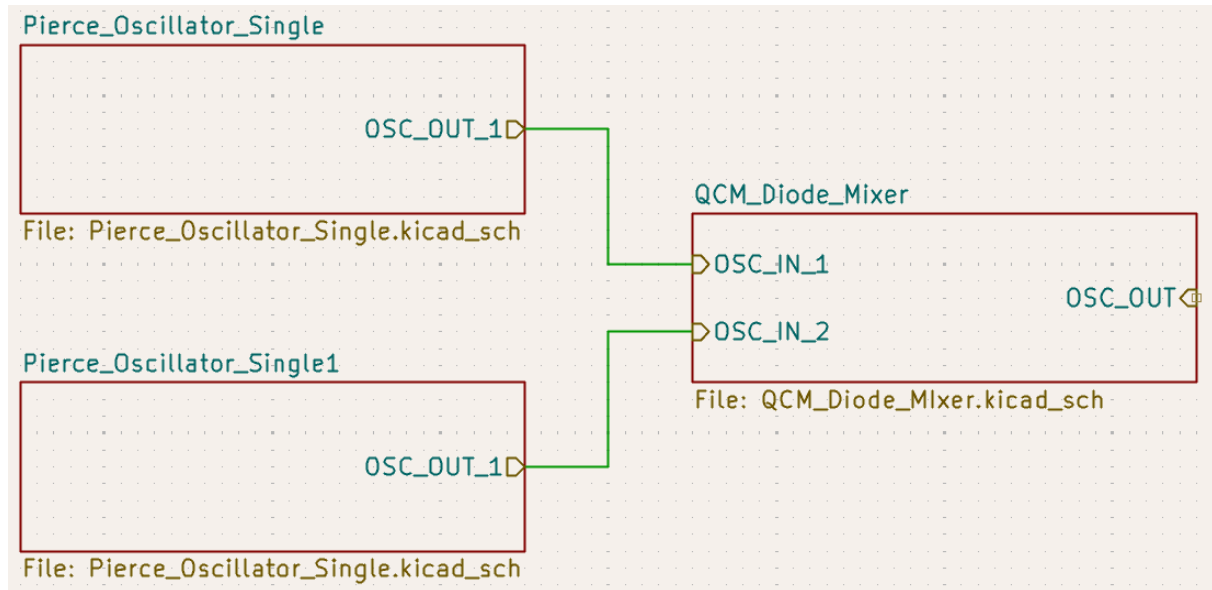
A Quartz Crystal Microbalance (QCM) measures microgram-level mass deposition during sputtering by tracking shifts in a quartz crystal's resonant frequency. To compensate for frequency drift caused by thermal fluctuations inside the chamber, a second reference crystal exposed to the same temperature is used. The heterodyne circuit mixes the sensing crystal's frequency with the reference crystal's frequency down to a low-frequency beat signal, which an STM32 microcontroller can precisely measure.

V1 Proof of Concept



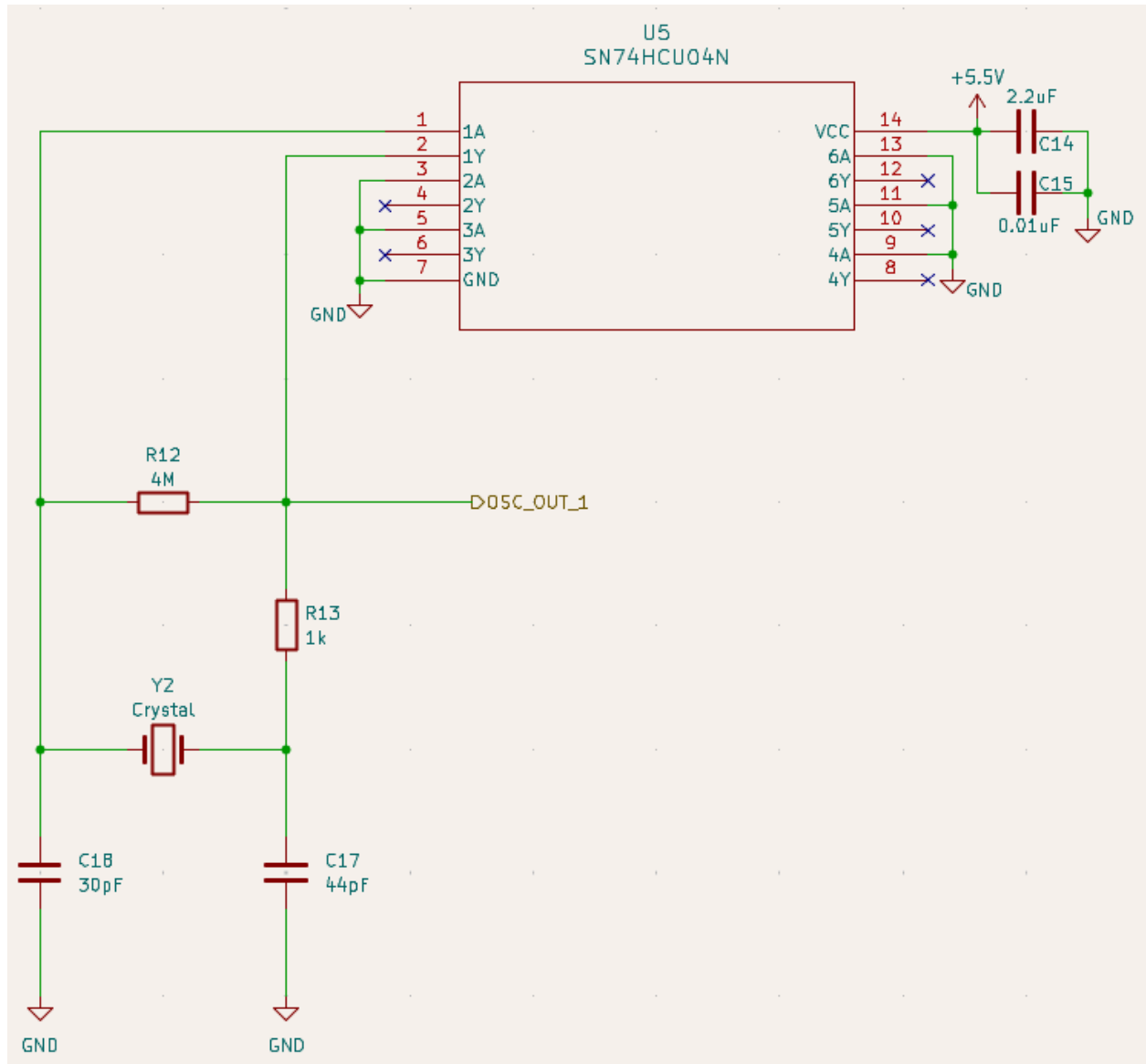
Pierce oscillator perfboard with custom 3D-printed quartz crystal fixture, connected to an STM32 Nucleo for frequency counting and validated using oscilloscope probes. Tests were conducted to measure sensitivity, noise and frequency fluctuations. It was then decided to move to a dual pierce oscillator to better compensate for thermal drift.

V2 - High level Architecture



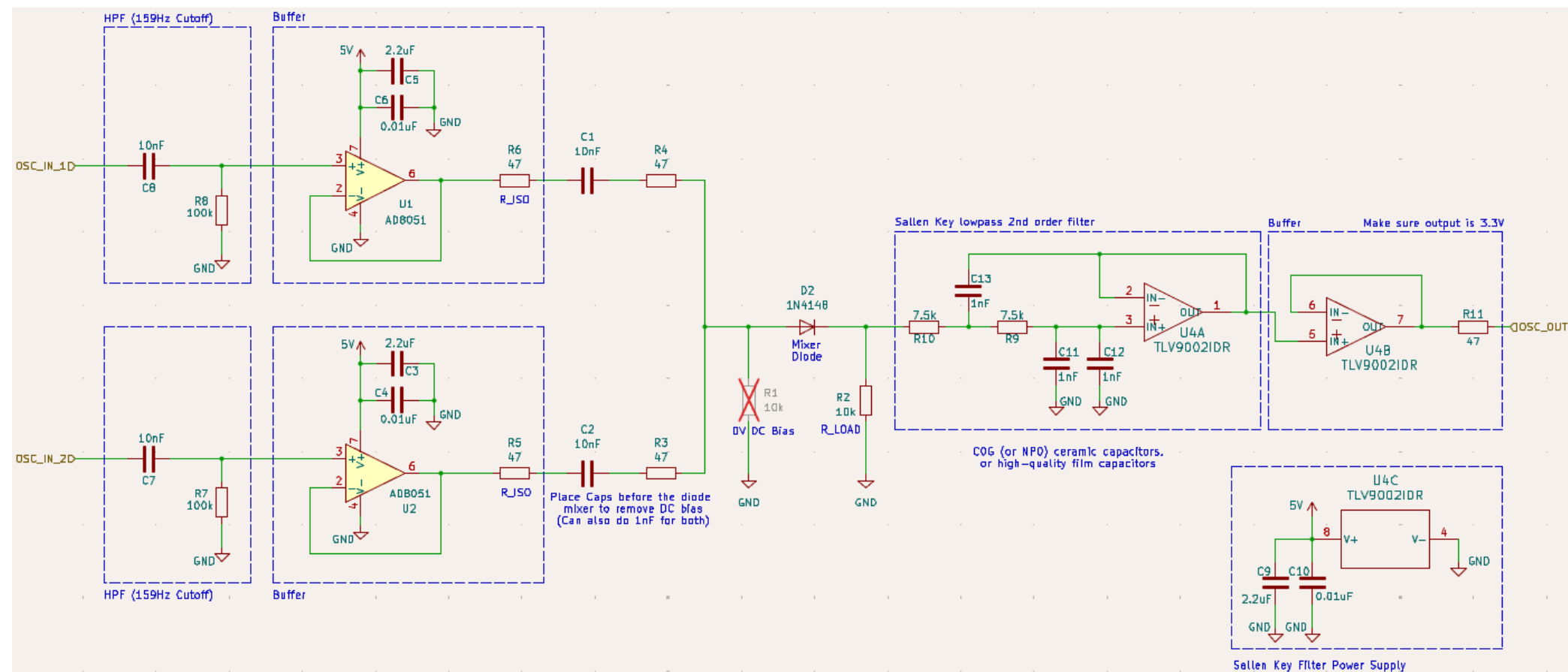
Hierarchical schematic in KiCad showing the dual Pierce oscillators (one reference and the other for measuring) feeding into a mixer stage for frequency difference extraction.

V2 - Pierce Oscillator Circuit



Schematic of the single-ended Pierce oscillator designed around an unbuffered hex inverter.

V2 - Heterodyne Mixer, Active Filter Stages, Buffer Stages



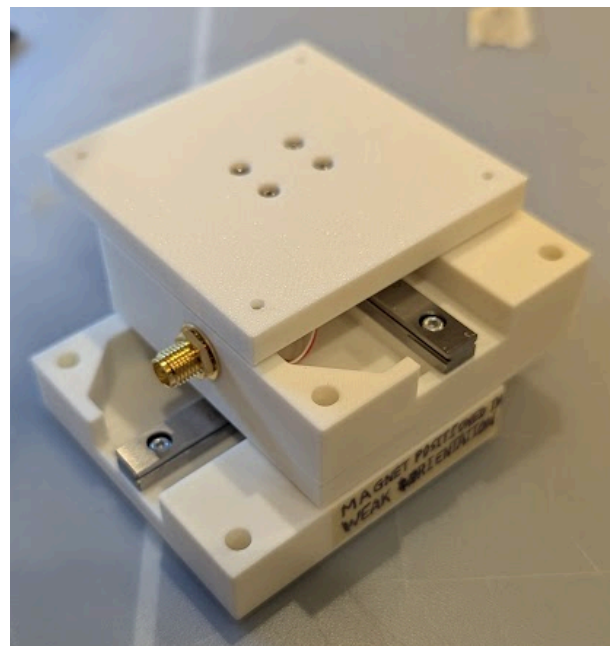
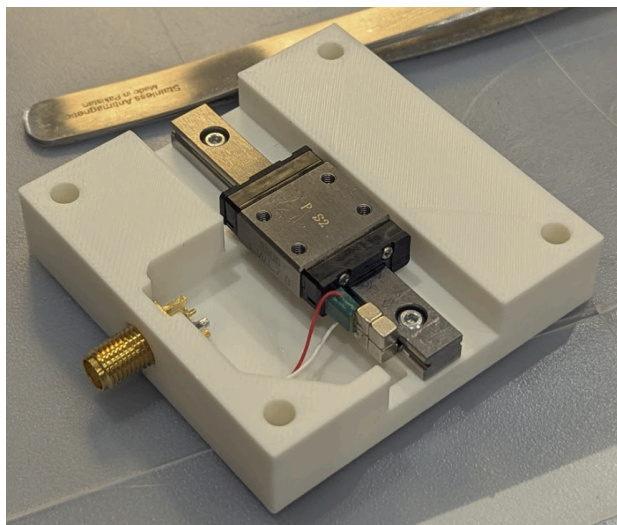
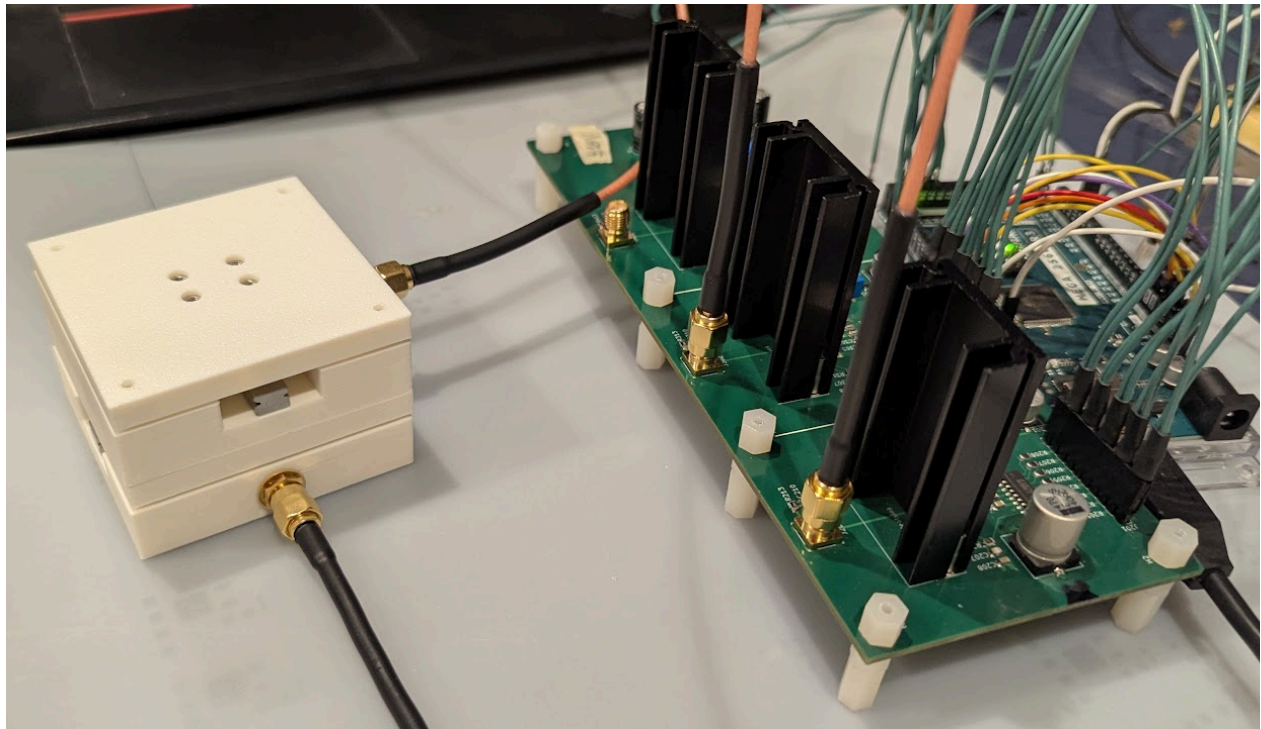
Features high-speed AD8051 input buffers, a 1N4148 diode rectification stage, and a 2nd-order Sallen-Key low-pass filter with a 15kHz cutoff (TLV9002) to isolate the low-frequency beat signal (Δf).

Status: KiCad design complete; undergoing bench validation. Future work includes comparative performance testing against XOR-gate and analog multiplexer mixer topologies.

Piezo-nanopositioner wafer stage (Hacker Fab team)

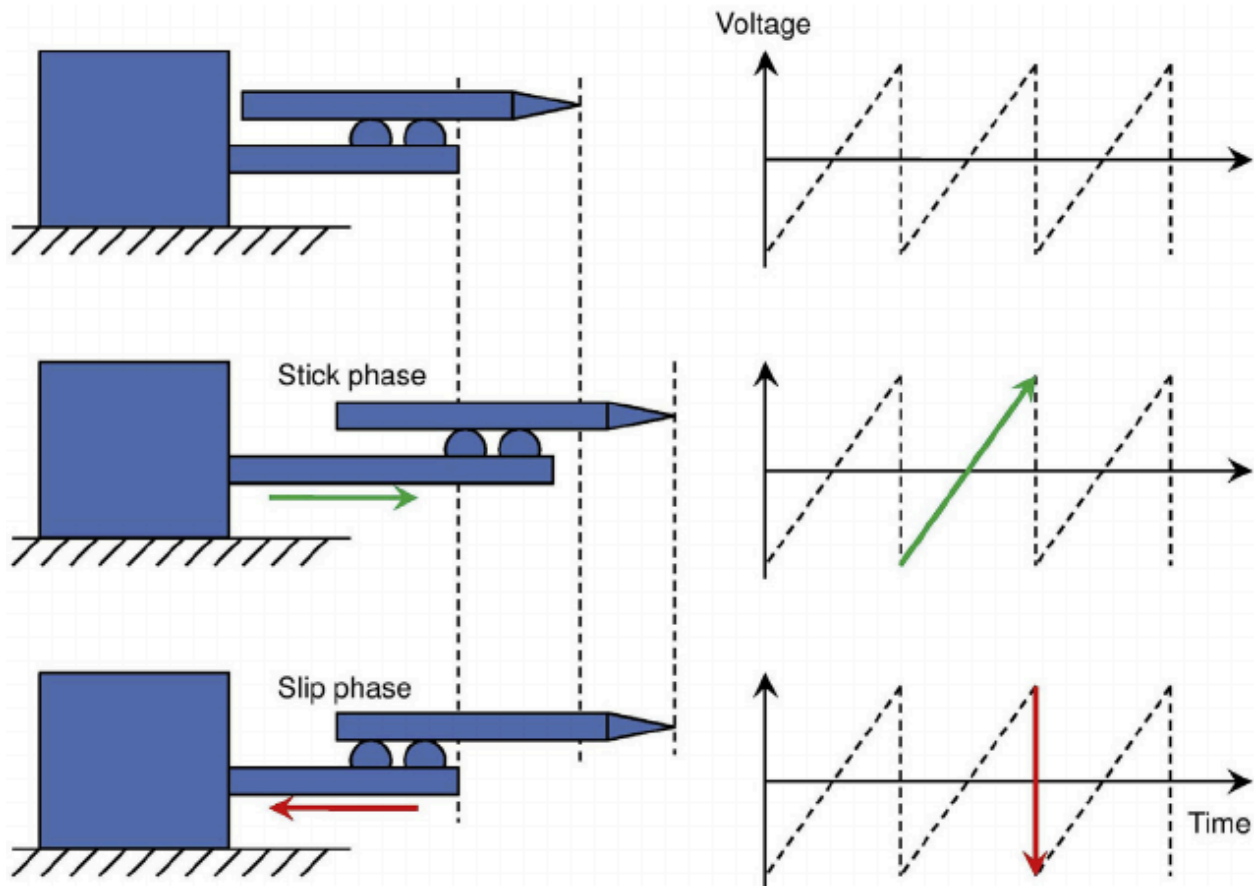
Project Description

Designed and built a 3-axis sub-micron wafer positioning stage utilizing a stick-slip piezo actuation mechanism for semiconductor lithography setups.



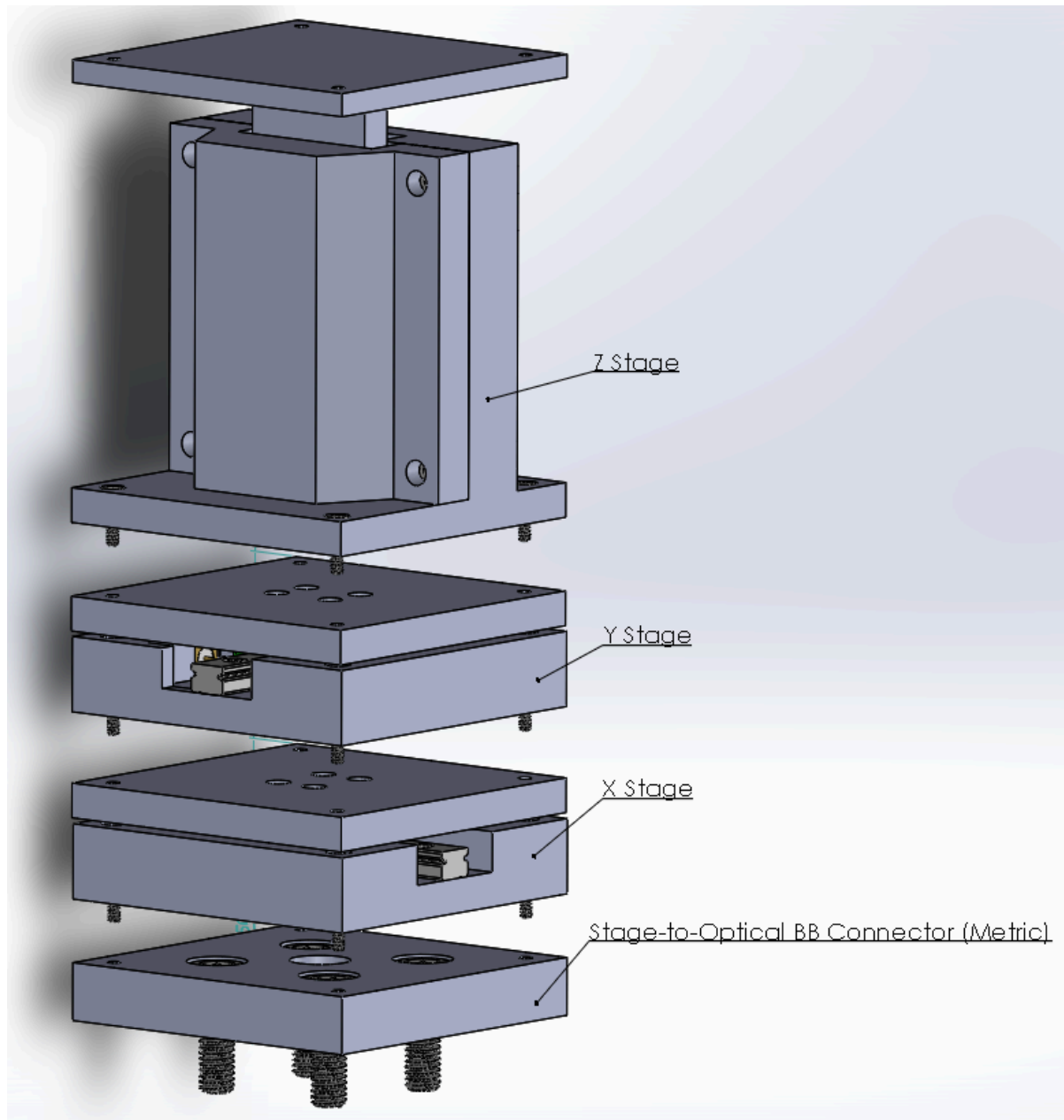
Stick slip mechanism description

Slow voltage ramp "sticks" the stage to move it forward; rapid voltage drop "slips" the drive element back without disturbing stage position.

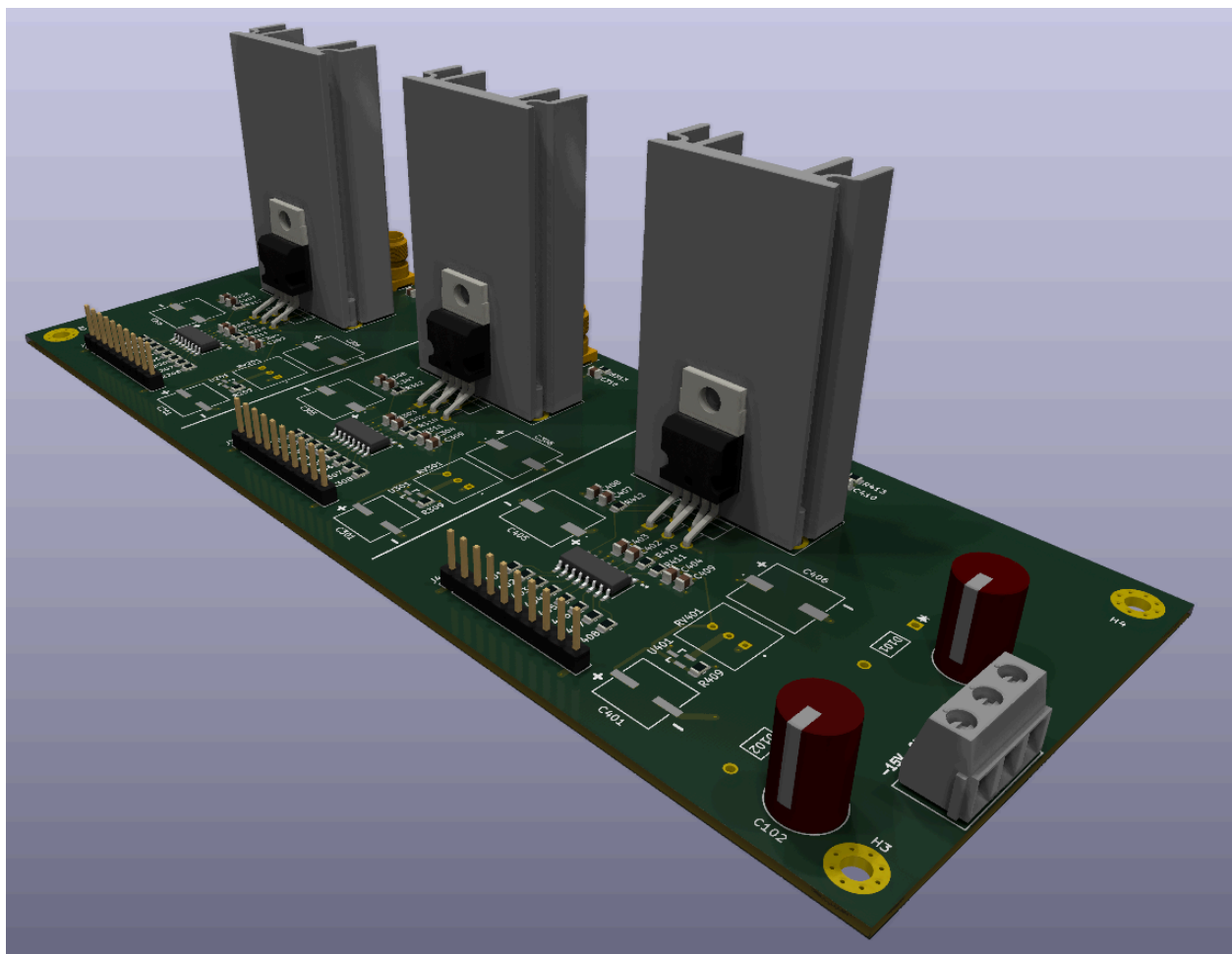
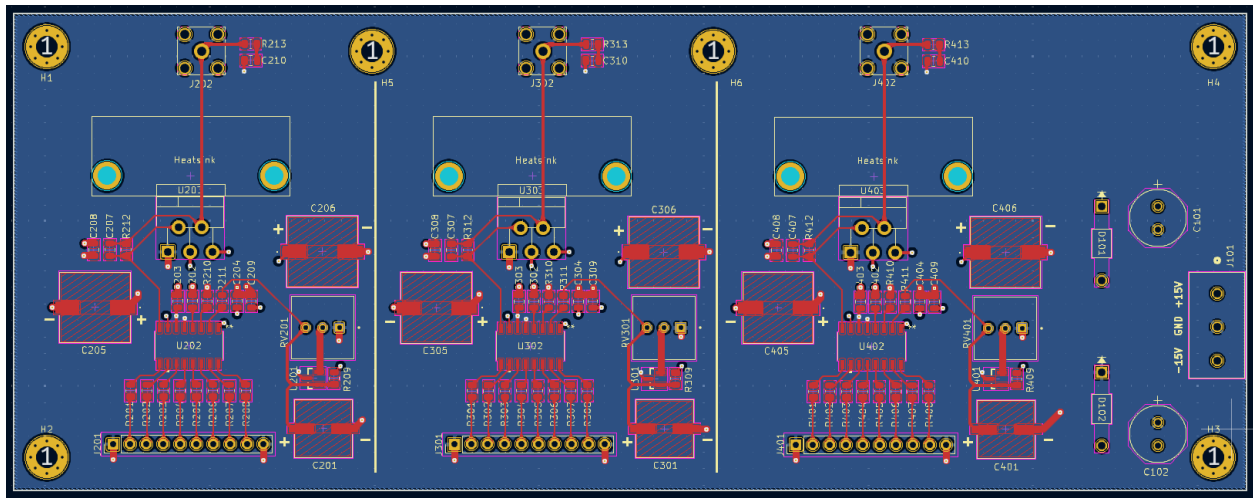


Mechanical & CAD

SolidWorks design featuring X, Y, and Z stages mounted to an optical breadboard.

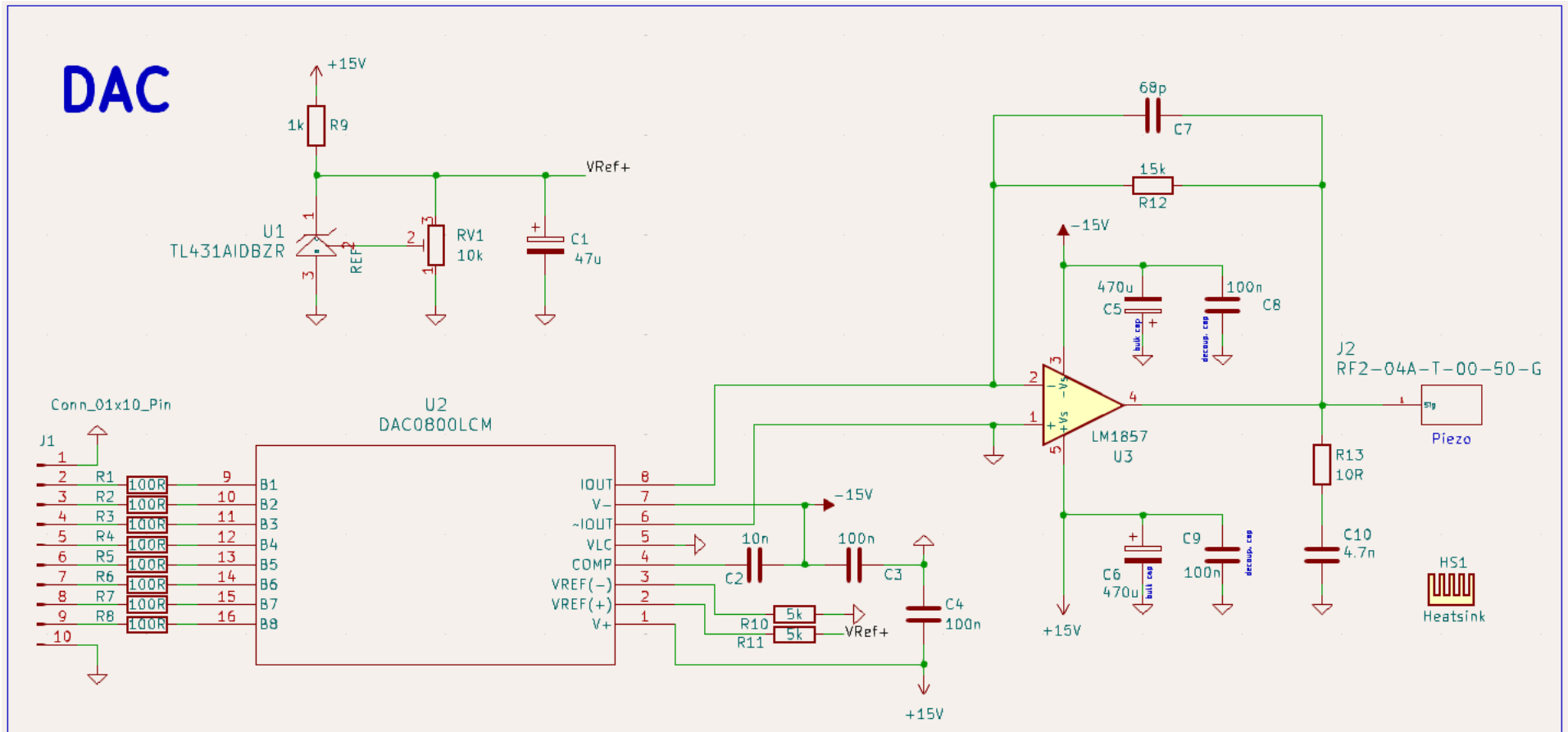


PCB Layout Design



Schematic Design

Used an 8-bit current-output DAC (DAC0800) feeding directly into an LM1875 power op-amp configured as a transimpedance amplifier (TIA) to convert output current into 15V piezo driving signals. The 15V rail was selected to achieve a 1 μ m piezo actuator displacement, however due to the magnetic preload force the net step size will be sub 1 μ m.



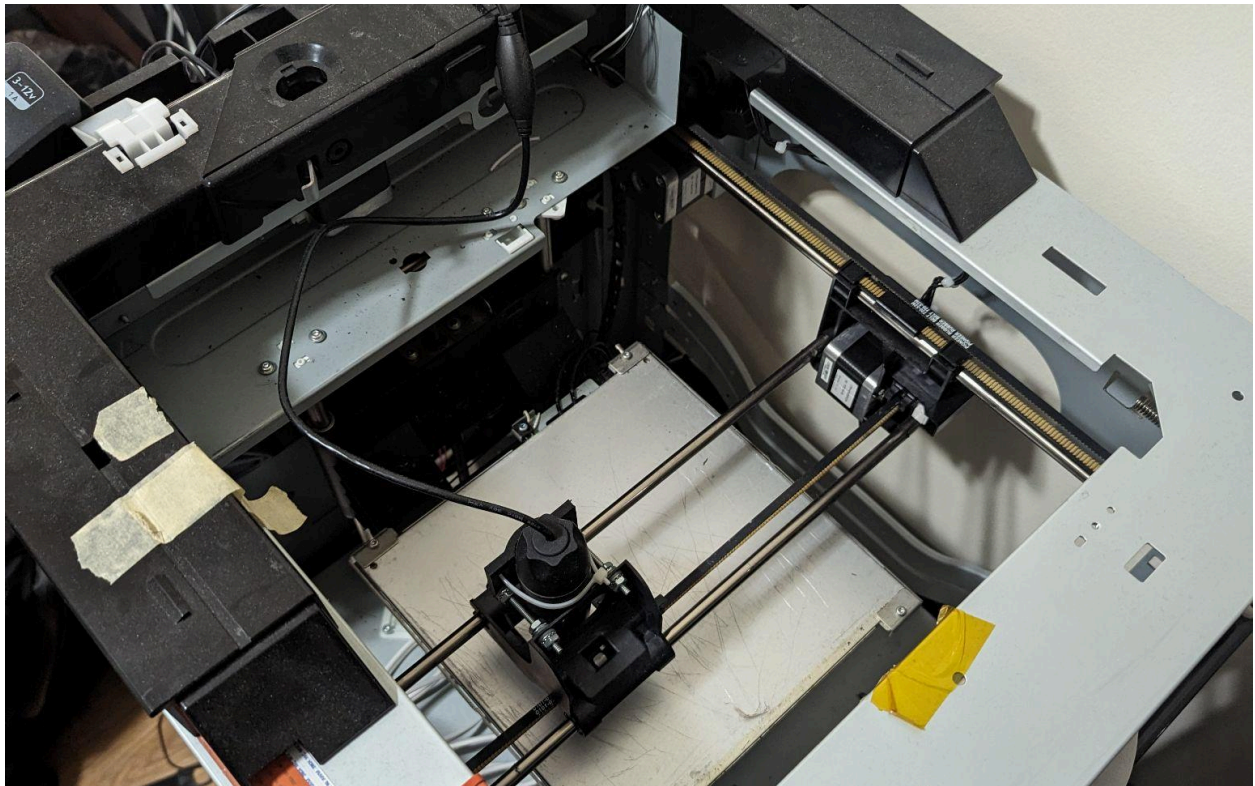
Github: <https://github.com/Waterloo-Hacker-Fab/Hardware/tree/master/Piezo-Nanopositioner-V1>

PCB Defect Detection Robot

Project Description

- Make an open source platform for creators/tinkers at home who want to produce and visually inspect their PCBs for defects at a small-scale manufacturing level.
- Attempted to solve this problem by using a combination of cheap hardware which can easily attach/dismount to a 3D printer.

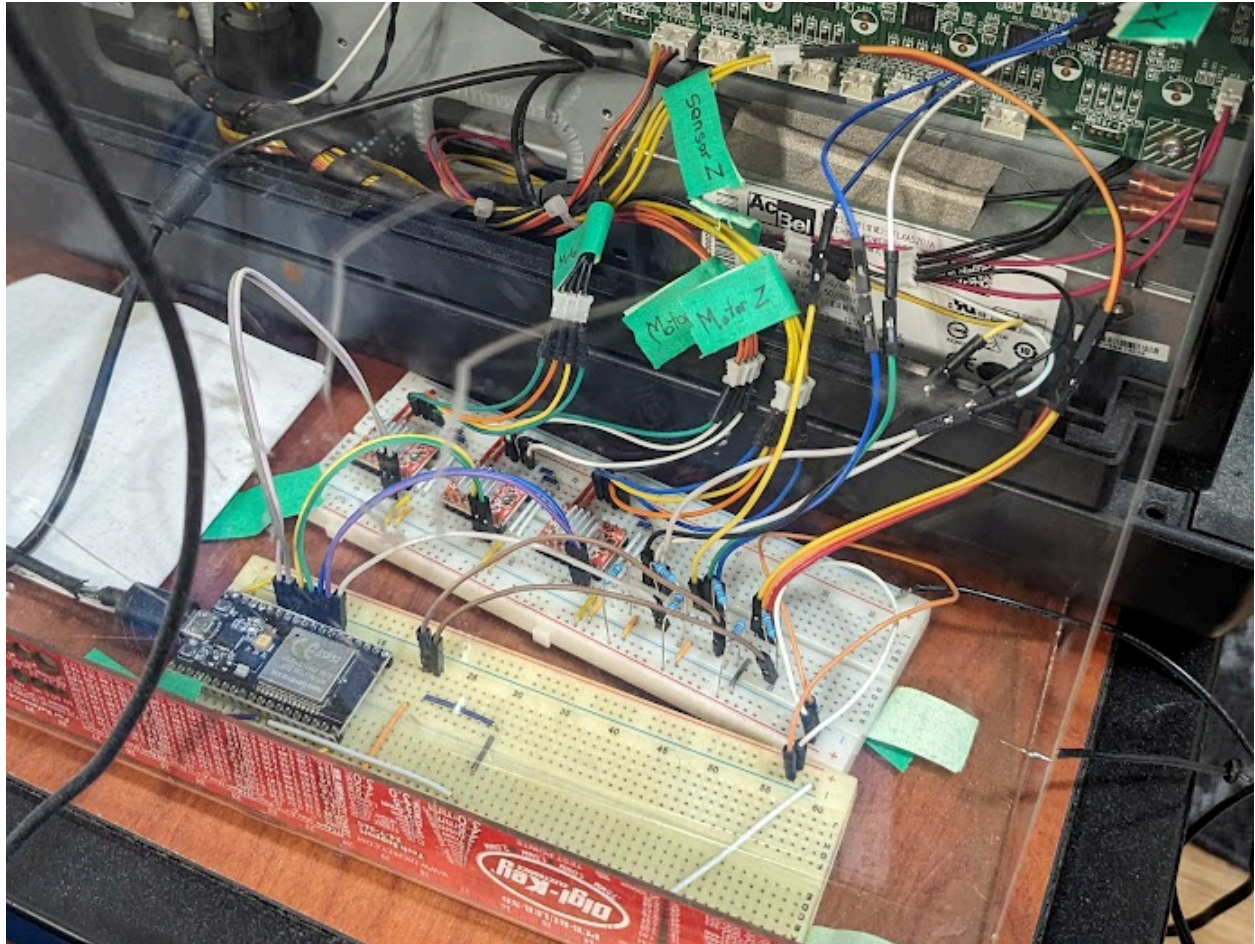
Picture of the head of the 3D printer where the microscope camera now replaces the nozzle of the printer.



Breadboard Prototype & Initial Testing

Motion Control: Driven via ESP32 running custom C++ firmware and three A4988 stepper drivers for XYZ motion.

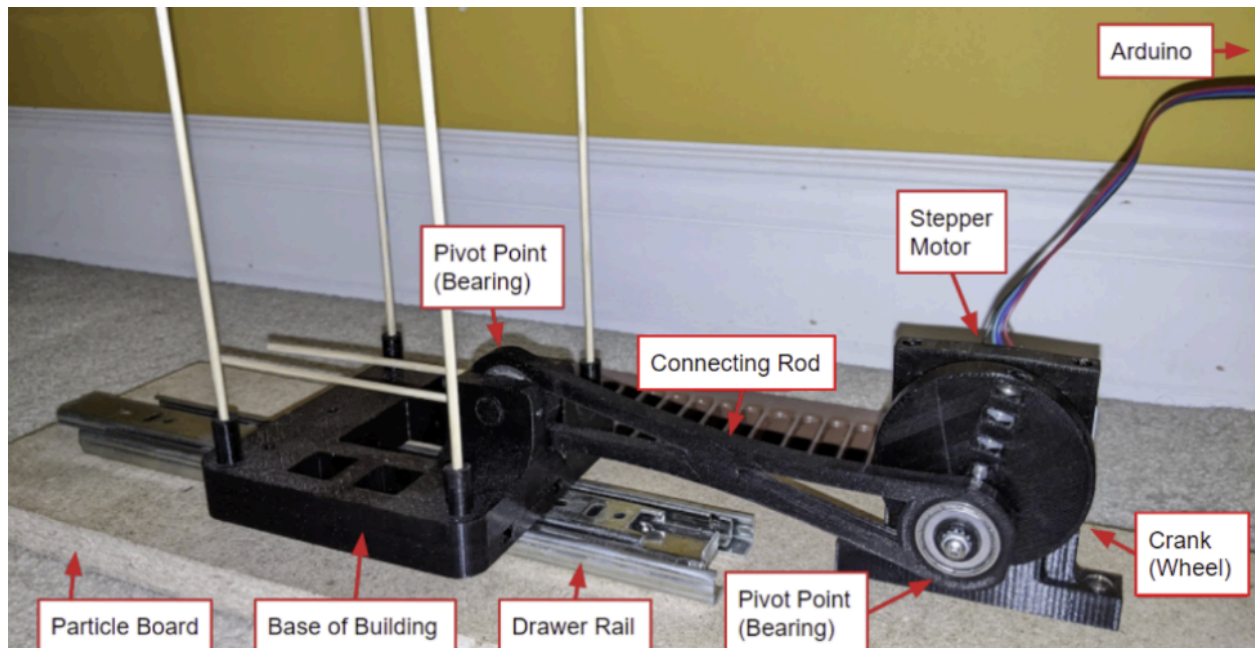
Wireless Interface: Implemented MQTT over WiFi to send real-time axis movement commands to the printer rig.

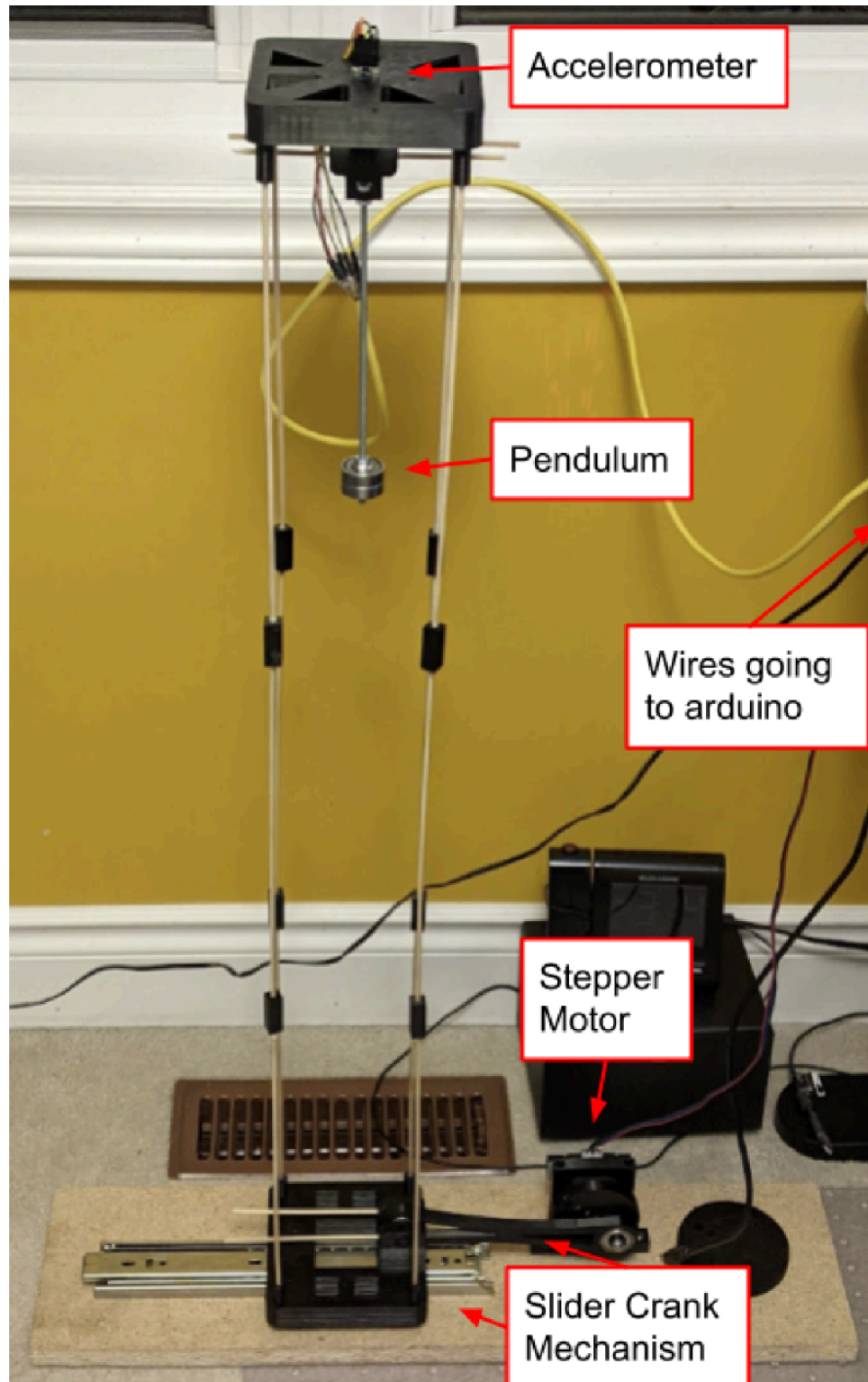


Tuned Mass Damper system experiment

Project Description

- The objective was to evaluate the effectiveness of pendulums for damping vibrations in buildings (from earthquakes).
- Built an automated slider-crank shake table driven by a stepper motor to induce controlled mechanical oscillations in a scale structural frame.
- Integrated a top-mounted accelerometer and Arduino (C++) to log real-time structural response and damping rates.
- Designed all custom mechanical components using SolidWorks.





Final test setup design

Electric Generator Experiment

Project Description

- Built an automated test fixture to analyze electromagnetic induction and evaluate generator performance.
- Designed custom mounting components in SolidWorks and 3D printed structural parts for the rotor and coil assembly.
- Programmed an Arduino (C++) and A4988 stepper driver to precisely control magnet rotation speed and streamline data collection.

